STANDATA variance 23-BCV-011

Building

Polyethylene film for greenhouse enclosures

Date Issued: February 2025

Purpose

This variance allows the use of products specifically designed for commercial greenhouse facilities occupied only by greenhouse staff, where members of the public are not permitted access.

Discussion

Flame-resistant fabrics and films conforming to CAN/ULC-S109, "Standard Method for Flame Tests of Flame-Resistant Fabrics and Films" are designed to be used for tents and air-supported structures within the scope of the National Building Code - 2023 Alberta Edition (NBC(AE)). Use of such materials for commercial greenhouses would be detrimental to plants due to the material's ability to block out the majority of the ultraviolet light necessary for plant growth.

For commercial greenhouse facilities, several types of polyethylene films with special additives for greenhouse use are available but do not meet the CAN/ULC-S109 criteria.

Code References

Article 3.1.18.5.

3.1.18.5. Flame Resistance

1) Every tent and *air-supported structure* and all tarpaulins and decorative materials used in connection with these structures shall conform to CAN/ULC-S109, "Standard Method for Flame Tests of Flame-Resistant Fabrics and Films."

Application

This variance applies to commercial greenhouse facilities not intended for the public.

Variance

This variance provides approximately equivalent or greater safety performance with respect to persons and property as provided for by the *Safety Codes Act* and the (NBC(AE)).

Polyethylene film used to enclose commercial greenhouse facilities is not required to conform to CAN/ULC-S109, "Standard Method for Flame Tests of Flame-Resistant Fabrics and Films," under the following conditions:

1. the greenhouse is used only for the growing of plants, (flowers, vegetables, shrubs, trees, etc.),
2. members of the public do not have access to the greenhouse, and
3. the polyethylene film has a flame spread rating not exceeding 25.

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For greenhouses used for other purposes, including facilities which include a sales area to serve customers, the application of CAN/ULC-S109, for enclosing materials would apply as per Article 3.1.18.5.

This VARIANCE is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are to Division B of the National Building Code - 2023 Alberta Edition

Issued by the Provincial Building Administrator

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